

DisasterSociety: Generative Agent-Based Simulation of Resident and Household Responses to Dynamic Urban Disasters

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Abstract—Urban disaster response is a dynamically coupled social process in which residents may receive different information, coordinate evacuation plans within households, and adapt their actions under shared constraints such as roads, vehicles, and care responsibilities. Generative agents can produce context-sensitive decisions and communication, while credible social simulation requires an explicit distinction among residents’ private cognition, household coordination commitments, and authoritative world state. We present *DisasterSociety*, a generative agent-based social simulation framework that maintains these states separately and connects them through explicit message interaction, commitment validation, world arbitration, and execution feedback. Resident language models observe, remember, plan, communicate, and propose structured action intentions, while household and world layers determine whether social agreements and physical actions are valid and executable. In a Carr-Fire-informed controlled scenario, paired ablations of memory, feedback, planning, and interaction show that all four mechanisms operate as intended at their interfaces, whereas only feedback and interaction exhibit consistent downstream effects. Removing feedback increases closed-route rejections by 6.542 per household; removing interaction eliminates coordinated multi-adult departure while individual evacuation remains possible. Cross-model and prompt-wording studies further show that these two mechanism effects are broadly preserved across decision models and instruction formulations. These results demonstrate the importance of evaluating both whether a generative-agent mechanism functions and whether it materially changes household- and collective-level social behavior.

Index Terms—Generative agents, multi-agent simulation, urban disaster response, household coordination, social simulation, event-driven simulation, mechanism ablation.

I. INTRODUCTION

Urban disaster response is a temporally evolving social process. Protective-action research shows that residents move through multiple stages, from receiving and interpreting warnings to assessing threats and protective actions and eventually acting under practical constraints [1], [2], [3], [4], [5]. These processes also unfold within households and social relationships. Household members may begin in different locations, receive different information, and coordinate gathering, shared-vehicle use, care responsibilities, and departure timing; their responses may ultimately take the form of joint or split evacuation [6], [7].

Fig. 1 illustrates the type of coupled process considered in this work. Resident A has received an official warning

whereas Resident B has not, leaving the two residents with different private cognitions. Through communication, they negotiate a driver, companions, route, and departure time and form a validated household commitment. When the primary route later closes, the original action is rejected by the world, and the execution outcome supports replanning toward the alternate route. This process highlights three requirements for disaster social simulation: representing how information becomes available to individual residents, how household members form shared coordination states, and how socially formed intentions become actions under physical constraints.

Agent-based models provide a bottom-up representation of heterogeneous residents and their interactions, while large language models (LLMs) further support context-sensitive judgment, planning, and natural-language communication. Generative Agents connected memory, reflection, planning, and interaction in a persistent artificial society [8], and subsequent work has extended LLM-agent simulation to larger societal and urban settings [9], [10], [11]. Disaster-oriented research now includes theory-informed individual evacuation prediction [12], LLM residents embedded in spatial fire and crowd environments [13], [14], and integrated simulations connecting hazards, warnings, and populations [15]. These developments substantially expand generative resident modeling, while explicit representations of household-level shared coordination and the transition from socially formed intentions to physically executable actions remain limited.

We formulate this challenge in terms of three state boundaries that must be maintained in disaster social simulation: resident-private cognition, household-shared commitments, and authoritative world state. Co-residence does not automatically synchronize warnings, memories, route beliefs, or private plans across household members. A message that has been sent, delivered, or even accepted becomes a household commitment only after member, resource, and timing constraints are satisfied. Likewise, an action intention produced by a language model must be arbitrated against a common physical snapshot before roads, vehicles, locations, and competing resources can be consistently updated. A generative social simulation therefore requires explicit transitions from world information to private cognition, from private proposals to shared coordination, and from shared plans to physical execution.

This leads to the central research question of this paper: *how can LLM-driven residents form information-bounded, traceable, and executable social behavior processes in a dy-*

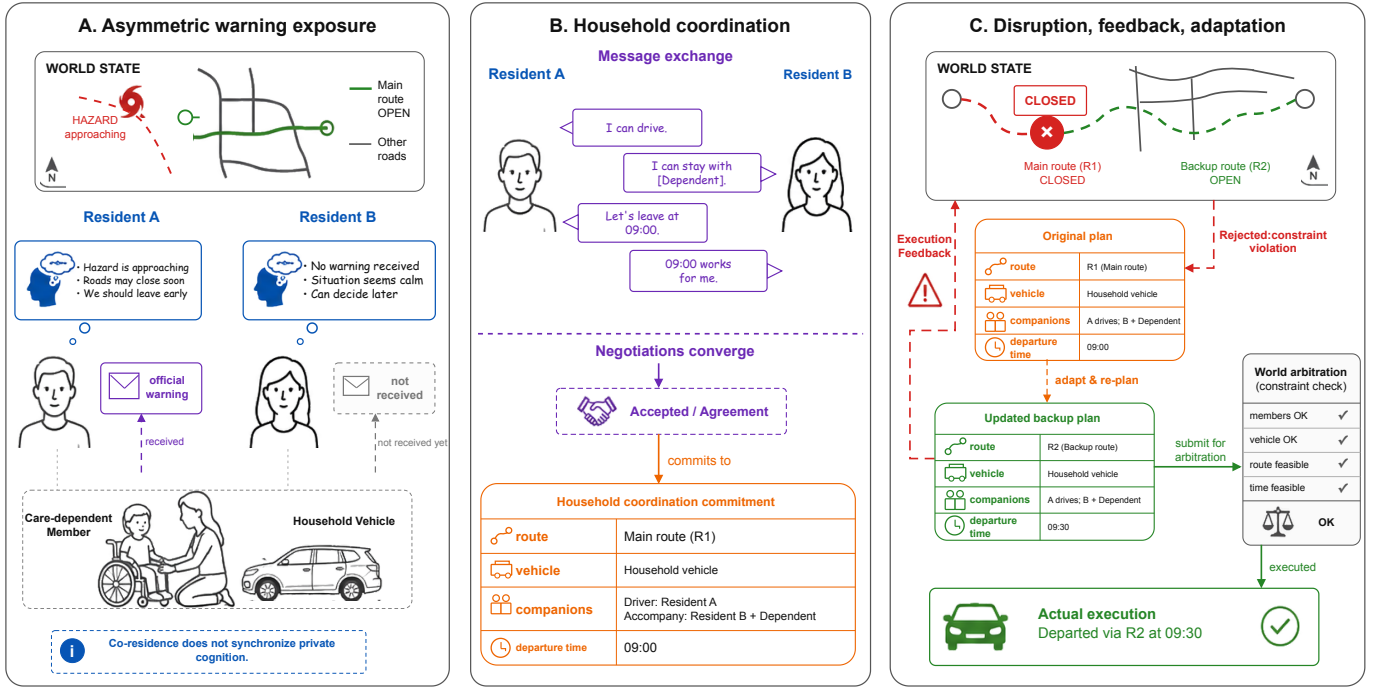


Fig. 1. Running example of a coupled household evacuation process. (A) Asymmetric warning delivery gives co-resident adults different private beliefs while the household still shares a vehicle and care responsibility. (B) Message exchange converges to a validated household commitment recording route, vehicle, companions, and departure time. (C) Closure of the primary route R1 causes world arbitration to reject the plan; execution feedback triggers replanning to the open backup route R2, and the revised action executes. The example is Carr-informed and controlled; it illustrates the modeled mechanisms rather than reconstructing the historical Carr Fire timeline.

namic, constrained socio-physical environment? Information boundedness requires residents to act only on information obtained through permitted observation, delivery, and feedback channels. Traceability requires explicit coordination records connecting private messages to shared household plans. Executability requires every action that may change shared resources or physical state to pass deterministic validation.

To address this question, we present DisasterSociety. The framework maintains resident-private cognition, household-shared state, and authoritative world state separately and connects them through explicit social-interaction and action-execution processes. Resident language models use bounded observations to remember, plan, communicate, and generate structured action intentions. Messages progress through sending, delivery, processing, and response before accepted proposals can become household commitments under explicit constraints. Action intentions are then batch-arbitrated against one authoritative world snapshot, and executed or rejected outcomes are returned to resident cognition. Together, these transitions form a traceable closed loop from private cognition to household coordination and physical execution.

This paper makes three contributions. First, we introduce a state-ownership and execution architecture for generative disaster social simulation that represents resident-private cognition, household-shared constraints and commitments, and authoritative world state as three separately maintained state spaces connected by explicit transitions. Second, we develop a closed social-process model linking communication and action through a coordination chain

(sent → delivered → processed → accepted → feasible commitment) and an execution chain (intention → world arbitration → executed/rejected → feedback), yielding an explicit path from private cognition to household coordination and physical action. Third, we introduce a mechanism-oriented evaluation design that separately tests whether memory, feedback, planning, and interaction operate at their intended interfaces and whether their changes propagate to household- and collective-level outcomes, with paired ablations, cross-model replication, and prompt-wording sensitivity used to assess the stability of the resulting mechanism findings.

The Carr-Fire-informed controlled case serves as the experimental setting for studying these social processes and mechanisms. Section II reviews related work. Section III presents the state model, social-coordination mechanisms, and execution process. Section IV reports the process, ablation, and robustness studies. Sections V and VI discuss the findings and their boundaries and conclude the paper.

II. RELATED WORK

A. Protective Action and Household Evacuation Behavior

Protective-action research treats disaster response as a staged process: cues and warnings must be received, attended to, and understood before threat and action appraisal, which remain subject to situational impediments [1], [2], [4]. Warning exposure, preparation, and departure unfold over time rather than collapsing into a single binary outcome [5]; crisis-informatics fieldwork shows that official messages reach residents unevenly and are reinterpreted through social

channels before being acted on [16]. The process is also social at the household level: members distributed across locations coordinate gathering and shared-vehicle use, and observed plans do not always place all members in one departure group [3], [6], [7]. This literature specifies which social processes a resident-level model must represent; it does not dictate a software state machine or establish that an LLM reproduces those processes.

B. Agent-Based Disaster and Evacuation Simulation

Classical evacuation modeling is strong where generative cognition is weak. Dynamic traffic models represent departure-time choice, route choice, and network assignment at varying behavioral detail [17], and synthetic-population pipelines match joint household and person attribute distributions so that simulated cohorts remain demographically coherent [18], [19]. Belief–desire–intention evacuation models separate belief, intention, and execution in software, but the resident decision unit is rule-based rather than generative, and household commitments are not modeled as versioned shared objects [20]. This literature also supplies the evaluation standards we adopt: verification and validation as distinct activities [21], [22], pattern-oriented modeling [23], and transparent documentation [24]. Recent evacuation-guidance work in this venue varies scenario difficulty and benchmarks competing controllers [25]. DisasterSociety is upstream of network simulators: it produces the departure parties and route intentions a traffic model would then execute, and does not replace them.

C. Generative-Agent Social Simulation

Generative Agents combined episodic memory, reflection, planning, and communication in a persistent artificial community [8]. Subsequent platforms scaled LLM-agent simulation to societal and urban settings, in some cases through hybrid LLM/numerical designs to control token cost [9], [10], [11], and generative AI has been argued to augment social-science measurement and simulation alike [26]. Within this venue, social-network simulation frameworks formalize how individual behaviors aggregate into community-level dynamics [27], and organized multi-agent teams show that explicit leadership structure reduces communication cost in LLM cooperation [28]. In the systems reviewed here, agent communication is often represented without a separately validated household-level commitment state; an authoritative physical world that arbitrates whether a linguistically formed plan can execute is rare, and mechanism-linked ablation of resident cognition is rarer still.

D. LLM Agents in Disaster Response

LLMs have entered the disaster pipeline at several points. FLARE uses behavioral theory and LLM reasoning for individual wildfire evacuation prediction on post-event surveys [12]; Dang et al. embed LLM agents in a cellular-automata fire environment [13]; Yang et al. couple LLM decisions with pedestrian and vehicle evacuation in a dynamic ABM [14]; and RESPOND joins flood forecasting, alerts, road closures, and

population agents in an integrated demonstration [15]. Most recently, hierarchical generative agents organize one resident’s sequential evacuation decisions into three cognition levels and calibrate against empirical human evacuation data [29]. These works organize cognition *within* a resident; our focus is the complementary boundary *between* residents and the world—how private cognitions produce household-level coordinated and physically executable behavior.

Across these lines of work, the unresolved systems question is how to preserve resident-private information, turn accepted messages into versioned household commitments, and arbitrate physical execution against one authoritative world state. DisasterSociety makes these boundaries explicit and evaluates their mechanisms through controlled ablation (Sections III and IV).

III. METHOD

DisasterSociety is a discrete-time, event-driven generative-agent social simulation framework for studying how private resident cognition develops into household coordination and subsequently into physically constrained action under a changing disaster environment.

The framework maintains three separately owned persistent states: *resident-private cognition*, *household-shared state*, and *authoritative world state*. Resident language models form cognition, messages, and action intentions from information available through permitted channels. The household layer converts socially formed proposals into validated shared commitments, while the world layer evaluates contemporaneous action intentions against a common physical snapshot and resolves physical constraints and conflicts. Execution outcomes are subsequently returned to resident cognition and become part of the next decision cycle.

This separation assigns information acquisition, social coordination, and physical execution to explicit state owners and transition processes. Fig. 2 summarizes scenario initialization, the three persistent states, and the principal transitions connecting them.

A. State Model and Ownership Boundaries

Let residents be indexed by $i \in \mathcal{R}$ and households by $h \in \mathcal{H}$, with household membership defined by $g(i) = h$. At simulation step t , the system state is

$$S_t = (W_t, \{C_{i,t}\}_{i \in \mathcal{R}}, \{H_{h,t}\}_{h \in \mathcal{H}}, Q_t), \quad (1)$$

where W_t denotes authoritative world state, $C_{i,t}$ denotes the private cognitive state of resident i , $H_{h,t}$ contains the shared resources, constraints, and accepted coordination state of household h , and Q_t stores auxiliary event, receipt, message, and interaction-process state.

These states have independent ownership. A change to $C_{i,t}$ does not automatically propagate to another resident $C_{j,t}$. Private beliefs and plans enter household-shared state only through explicit social-coordination transitions, and neither household commitments nor private action intentions can alter physical state before world arbitration.

Consequently, members of the same household may hold different warnings or route beliefs; an unexpressed private plan

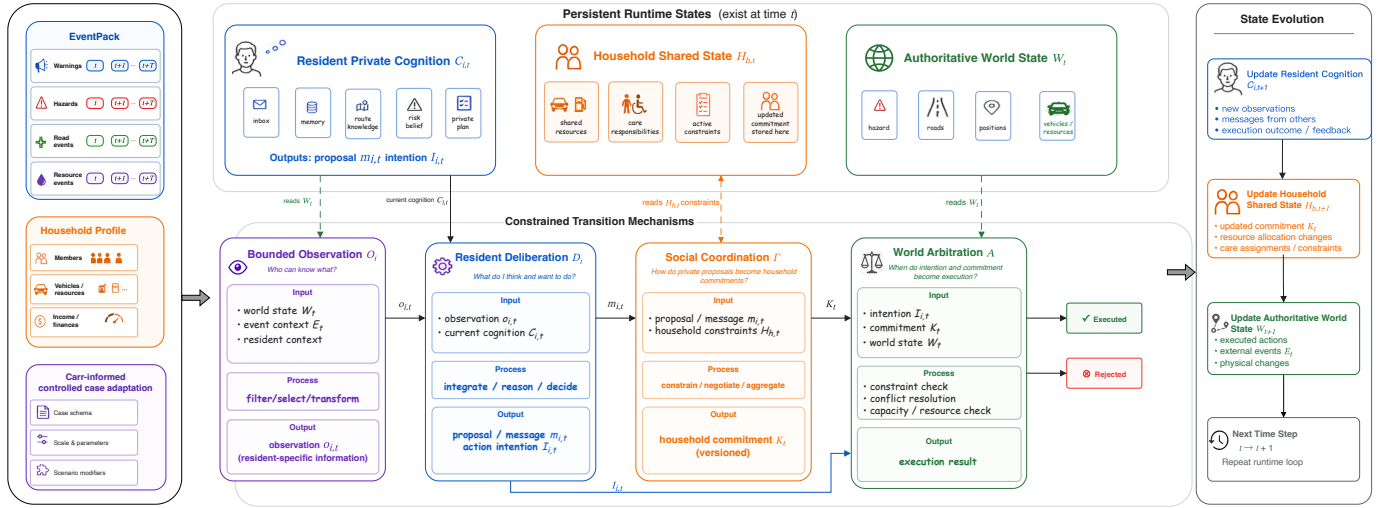


Fig. 2. DisasterSociety method framework. Scenario initialization combines a time-indexed *EventPack*, a household profile, and controlled case adaptation. The runtime maintains resident-private cognition $C_{i,t}$, household-shared state $H_{h,t}$, and authoritative world state W_t as separate persistent states. Bounded observation converts accessible context into resident-specific information $o_{i,t}$; resident deliberation produces a proposal or message $m_{i,t}$ and an action intention $I_{i,t}$; social coordination validates proposals into a versioned commitment K_t ; and world arbitration resolves $I_{i,t}$ and K_t against W_t , yielding an executed or rejected result. The updates become $C_{i,t+1}$, $H_{h,t+1}$, and W_{t+1} for the next step.

does not constitute a household agreement; and a resident's belief that a road is open does not alter its authoritative world state.

Resident i 's bounded observation at step t is defined as

$$O_{i,t} = \Omega_i(W_t, Q_{\leq t}; A_{i,t}), \quad (2)$$

where $A_{i,t}$ specifies information channels, recipient relationships, location, and adapter-specific access rules. The resident then produces a structured action intention

$$I_{i,t} = \Pi_i(C_{i,t}, O_{i,t}; \theta, \phi), \quad (3)$$

where θ denotes the declared decision model and ϕ denotes the enabled cognitive and interaction mechanisms. $I_{i,t}$ represents a behavioral proposal and does not itself modify household-shared or physical state.

At the implementation level, a *Resident* owns private cognition and decision state; a *Household* owns membership, vehicles, care responsibilities, and accepted household commitments; the *Interaction* layer owns message lifecycle state; and the *World* owns authoritative hazard, route, location, resource, and physical-arbitration state.

B. Information-Bounded Cognition and Structured Action Intentions

Resident decisions follow an explicit information boundary. Each resident may use only information obtained through permitted observation, message delivery, household relationships, and execution feedback. Accessible information may include the resident profile, currently visible household facts, delivered inbox items, privately perceived hazard and route state, selected private history, visible household commitments, and recent execution outcomes.

From this information, the language model returns a structured decision containing an assessment of the current situation, an action selected from a closed action set, an op-

tional destination, route, vehicle, departure time, accompanying members, outgoing messages, responses to previously received messages, and optional memory updates.

Structured outputs are validated for schema and identifier consistency before entering subsequent social-coordination or world-execution transitions.

C. Social Coordination, Household Commitments, and World Arbitration

Social interaction is represented through an explicit message lifecycle. Messages first progress through *sent* $\rightarrow$ *delivered* $\rightarrow$ *processed*, and may subsequently enter *accepted* or *rejected* states after an explicit recipient response.

These lifecycle states correspond to distinct social-process stages. Delivery determines whether a message enters the recipient's inbox, whereas processing determines when it becomes part of private history. Sending, receiving, and accepting therefore have different effects on subsequent cognition and coordination.

An accepted proposal becomes household-shared coordination state only after household-level validation. Let p denote a proposal for household h at step t . A household-valid proposal satisfies $\Phi(p, h, t) = 1$, with basic conditions requiring that the named decision members belong to the household, the proposed vehicle belongs to the household, a route is specified, and the proposed departure time is no earlier than the current step. Household commitments are versioned; when a new validated commitment revises an existing one, the previous version is cancelled while its supersession relation is retained.

A household commitment records social coordination, while physical execution remains subject to world arbitration. For contemporaneous action intentions, the World evaluates a common authoritative snapshot W_t and checks route availability and capacity, vehicle state, resident positions, care requirements, and competing resource claims.

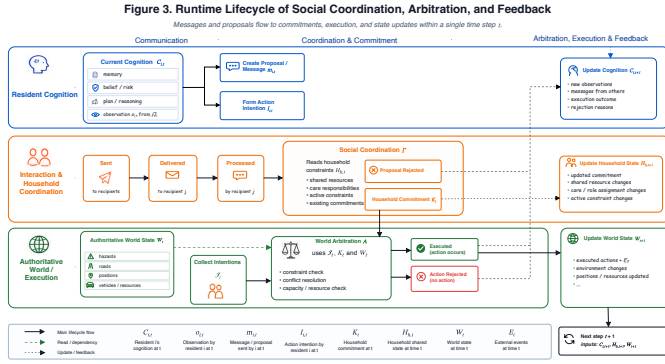


Fig. 3. Runtime lifecycle of social coordination, arbitration, and feedback within step t . Resident cognition produces a proposal or message $m_{i,t}$ and an action intention $I_{i,t}$. Messages move through sent, delivered, and processed states before social coordination either rejects the proposal or forms household commitment K_t . In parallel, collected intentions, commitment K_t , and authoritative state W_t enter world arbitration, which returns executed or rejected outcomes. Read dependencies and outcome feedback produce $C_{i,t+1}$, $H_{i,t+1}$, and W_{t+1} for the next step.

The world returns either an executed or rejected outcome. Only executed outcomes may change resident locations or physical resources. Rejected outcomes leave the corresponding physical state unchanged and retain a rejection reason that may enter subsequent resident cognition and planning.

DisasterSociety therefore links two processes, *message* $\rightarrow$ *household commitment* and *action intention* $\rightarrow$ *world arbitration* $\rightarrow$ *execution outcome*. The first describes how private communication forms household-shared coordination, and the second describes how socially formed action plans become physical behavior.

D. Event-Driven Process Loop and State Update

Each simulation step advances social and physical state through a fixed sequence. Fig. 3 summarizes the runtime relationship among resident cognition, message coordination, household commitments, world arbitration, and feedback.

Each step consists of six ordered phases. (i) *World advance*: update simulation time, hazard state, and due exogenous events, and generate associated official warnings or orders. (ii) *Wake*: identify residents that require a decision at the current step. (iii) *Decision*: solicited residents generate messages and action intentions in parallel against the same immutable world-state snapshot. (iv) *Interaction*: process current inbox items and recipient responses and update private cognition and household coordination state. (v) *Arbitration*: collect contemporaneous action intentions, resolve them against a common authoritative world state, and return execution outcomes. (vi) *Delivery and record*: advance the delivery state of newly emitted messages and persist the resulting step state and process records.

This ordering ensures that residents act only on information they have obtained, contemporaneous decisions face a consistent physical environment, rejected actions do not consume shared resources or change locations, and execution outcomes can enter cognition, planning, and coordination in subsequent steps.

At the end of the step, resident-private cognition, household-shared state, and authoritative world state are updated to $C_{i,t+1}$, $H_{i,t+1}$, and W_{t+1} , which jointly provide the state for the next simulation cycle.

E. Scenario and Population Instantiation

Exogenous disaster processes enter the simulation through an *EventPack* and scenario adapter. The EventPack organizes warnings, hazard changes, road events, and resource events, and the scenario adapter converts these inputs into time-indexed events consumed by the simulation kernel. Authoritative hazard and road changes are therefore defined by the external scenario.

The experiments use a Carr-Fire-informed controlled scenario. Carr Fire data provide contextual and provenance constraints. The final fire perimeter serves as a scenario reference, the two routes are derived experimental assets, and the road closure is introduced as a controlled perturbation. The resulting world state may therefore maintain hazard distance, route state and capacity, resident locations, vehicles, care obligations, and household coordination commitments.

The resident population is instantiated from weighted household microdata. Iterative proportional fitting (IPF) adjusts donor-household weights to match household-level margins for the target area [18], [19]. Sampling is then performed at the household level: when a donor household is selected, its member records are copied together and assigned new synthetic identifiers. This preserves internally observed relationships such as co-residence, shared vehicles, and care dependencies.

Initialization maps the sampled population into resident, household-resource, and care states. Members aged 18 years or older form the transparent decision-capable baseline, while minors are represented as dependent execution members. Vehicles maintain household ownership, location, capacity, occupancy, and reservation state, providing shared-resource constraints for subsequent household coordination and world arbitration.

IV. EXPERIMENTS

Three studies instantiate the claim families of our evaluation methodology: an empirical-process study under the full condition with the richer adapter, extended by a two-seed scale demonstration at 100 households (Section IV-B), the central mechanism ablation with paired full-versus-removal runs over four mechanism switches (Section IV-C), and a bounded cross-model replication of the two determined mechanism effects under seven decision models, together with an equivalent-prompt sensitivity check on the same effects (Section IV-D). All three run the Carr-informed controlled scenario of Section IV-A1; none is a historical reconstruction of the Carr Fire, and no reported quantity is a population-level evacuation prediction. Existing platforms do not expose matched mechanism controls; we therefore compare mechanism conditions within one platform, add a deterministic in-kernel reference policy (Section IV-C), and replicate the effects across decision-model families and sizes.

A. Experimental Design and Setup

1) *Two Experimental Operationalizations*: The shared kernel hosts different operationalizations of resident cognition, and the Carr case uses two adapters with deliberately different estimands (Table I): one platform, two operationalizations, two research questions. Both obey the same intention–outcome boundary and state separation; they differ in cohort construction, private-state representation, and the claims their artifacts support, and neither operationalization validates the other’s target.

a) *Empirical-process adapter*: This adapter studies a Carr-informed resident process under a full mechanism configuration with richer donor-microdata trait profiles. Residents are exposed to a modeled communication substrate (household cliques, nearby neighbors, and a friend ring—not observed Carr social ties), and coordination takes the form of structured departure parties whose member, caregiver, vehicle, time, and overlap checks run at acceptance, with route openness enforced by the world at execution. A valid solo-adult intention may also form a solo party without prior negotiation, so the adapter observes coordination where it occurs without forcing every departure through dialogue; solo parties are reported separately from multi-adult coordinated departures.

b) *Controlled-mechanism adapter*: This adapter supports paired mechanism identification: conditions share one household cohort, an initial private primary-route plan, a time-indexed disruption, and component-isolated exogenous streams, while four code-level flags enable or remove memory, feedback, planning, and interaction. Its defining feature for the ablation is the separation of private route perception from world route truth: with feedback enabled, a decision copies changed world route state into the private view and rejected execution adds bounded failure feedback; with feedback disabled, the pre-closure view persists while the world still arbitrates the true closure. Memory has a deliberately narrow operational meaning here—delivery-gated warning retention plus model-emitted memory writes—while the platform’s general ranked memory [8] remains available through the base resident interface. The remaining contract differences are summarized in Table I.

2) *Reproducibility Infrastructure: Gateway, Seeding, and Run Records*: Three properties of the execution infrastructure bear directly on the evidence reported in Section IV; the remaining engineering details are omitted. First, every language-model call passes through a single audited gateway that enforces schema validation, one repair attempt, and budgets; a failed call either aborts the run or enters a logged rule fallback that remains subject to world arbitration, and the fallback share is reported for every run. Second, controlled-scenario non-LLM randomness is drawn from component-isolated keyed streams that exclude the experiment condition label, so paired conditions share exogenous draws and a condition switch cannot masquerade as a mechanism effect. Third, each run writes step-level world, intention, outcome, and interaction records and a terminal summary, making every reported number recomputable from the persisted artifacts.

Evaluation is organized along two orthogonal dimensions. *Analysis scale* distinguishes individual, household or network,

collective, and system quantities. *Evidence type* distinguishes verification, behavioral reasonableness, mechanism-specific removal effects, and robustness checks. This separation follows established simulation methodology [21], [22], [23]; generative social simulation adds language-model and prompt dependence, motivating ablations and replication-risk checks [8], [26].

The Carr case uses two non-interchangeable populations. Carr-R is a version-fixed respondent table for bounded empirical references; Carr-S is a synthetic household sample for social-process experiments. No survey field is used both as a run input and as an evaluation target in the same experiment. Comparisons with Carr-R assess Carr-informed reasonableness rather than independent validation; the small non-evacuator group and absence of completed blinded review bound behavioral-validity claims.

For controlled mechanism evaluation, each module has a runner-aligned manipulation check: memory retains delivered warnings; feedback records a post-closure alternate-route update among residents with a pre-closure primary-route plan; planning completes a post-closure alternate-route replan; interaction forms an all-decision-member commitment referencing the alternate route, a household vehicle, and a departure step. These are scenario-specific checks, not general psychological-validity measures. Downstream quantities are computed from authoritative final state with distinct denominators. The household-level headline is the *complete safe household departure rate*: decision-capable members evacuated and dependents safe. Other quantities include decision-resident evacuation, closed-route rejections, multi-adult coordinated-departure rate, dependent safety, and shared-resource conflicts.

The controlled contrast for module m and paired seed s is

$$\Delta_{m,s} = Y_{s,\text{full}} - Y_{s,\text{full}-m}, \quad (4)$$

a conditional removal effect, not a full factorial decomposition. The run-level observation is a complete seed $\times$ condition run; paired cells share a seed and independent replication is across seeds. Seed-level differences are summarized by

$$\bar{\Delta}_m = \frac{1}{|S|} \sum_{s \in S} \Delta_{m,s}, \quad (5)$$

with paired- t intervals used only for descriptive visualization in Figs. 4 and 5. Sensitivity is reported through per-seed direction shares and exact-zero shares.

All model traffic passes through the audited gateway (Section IV-A2). No reported cell is excluded: all complete their planned steps with zero failed calls and zero fallbacks. Every reported quantity is recomputed from archived per-cell artifacts.

B. Do Households Complete the Response Process End to End?

This study exercises the full condition with the empirical-process adapter (Section IV-A1): 24 households carrying PUMS/Carr-informed trait profiles, a constructed social graph, recipient-specific official receipts, and structured departure

TABLE I

CARR ADAPTER CONTRACTS. BOTH ADAPTERS SHARE THE KERNEL, THE TYPED INTENTION–OUTCOME BOUNDARY, AND HOUSEHOLD/WORLD STATE SEPARATION, BUT DIFFER IN COHORT, PRIVATE-STATE REPRESENTATION, COMMUNICATION RECORD, AND ESTIMAND.

Contract	Empirical-process adapter	Controlled-mechanism adapter
Purpose and cohort	Carr-informed full-condition process study with richer PUMS/Carr trait profiles; profile ordering and formal cohort are fixed	Mechanism identification with seed-selected households containing exactly two adults, at least one dependent, and exactly one reported vehicle
Private cognition	Inbox, recent memory list, perceived routes, private commitment view, and execution feedback; no persistent executable plan	Mechanism flags, warning-history signal, feedback-gated route view, executable plan and plan history, accepted commitments, and recent failure feedback
Communication	Recipient-specific official receipts; social messages use a message-level record over a recipient list and a constructed social graph	Recipient-specific receipts; household direct and community-audience delivery are distinct, but the community audience is not a social-network topology
Coordination	Structured departure parties with care, vehicle, member, time, and overlap checks, plus an allowed solo auto-party path	Versioned controlled household proposals and commitments for the two decision members, with final execution checks in the world
Estimand	Empirical reasonableness and process-constraint correctness; social-message quantities reported at message level	Paired full-versus-drop-one mechanism contrasts with manipulation checks and downstream outcomes

TABLE II

EMPIRICAL-PROCESS OUTCOMES OVER 3 SEEDS (24 HOUSEHOLDS, 25 STEPS). RATES ARE SEED MEANS $\pm$ SD; MESSAGE, COMMITMENT, AND VIOLATION FIGURES ARE TOTALS OVER THE THREE RUNS. COMMITMENTS ARE REPORTED SEPARATELY FOR NEGOTIATED MULTI-ADULT COMMITMENTS AND SOLO AUTO-PARTIES, WHICH THE ADAPTER RECORDS AS DISTINCT COORDINATION PATHS RATHER THAN ONE FUNNEL.

Quantity	Value
Complete safe household departure rate	0.514 ± 0.024
Coordinated-departure household rate	0.333 ± 0.042
Split-departure household rate	0.042 ± 0.020
Messages sent $\rightarrow$ deliv. $\rightarrow$ processed	1214 $\rightarrow$ 710 $\rightarrow$ 665
Messages accepted / rejected / unanswered	467 / 73 / 125
Departure parties executed (per seed: 10 / 15 / 13)	38
Negotiated commitments accepted (all executed)	15
Solo auto-parties executed	23
Order receipts per run (vol. / mand.)	5 / 15
Care, vehicle, capacity, caregiver violations	0

parties over 25 steps. Three seeds (7201, 8301, 9401) complete all planned steps successfully. The decision model is DeepSeek-V4-Flash, and no call fails or falls back to the rule policy.

Table II summarizes the run-level outcomes. Roughly half of the households reach a complete safe household departure (0.514 ± 0.024 across seeds; per-seed values 0.500, 0.542, 0.500), one third depart as a coordinated multi-adult party, and split departures are rare. The message funnel retains every life-cycle stage—sent, delivered, processed, accepted—so the lossy channel, not a broadcast assumption, shapes coordination. Execution is similarly clean: 38 departure parties execute across the three runs with no party rejected at execution time. The coordination record keeps the two formation paths distinct: 15 negotiated multi-adult commitments are accepted and all execute, while the other 23 executed parties are solo auto-parties formed directly from a valid solo-adult intention without prior negotiation (Section IV-A1); neither path records a rejection or cancellation. Across all runs, world arbitration records zero care left-behind, vehicle-conflict, capacity, and caregiver violations: every executed departure party satisfied the household and resource contracts of Section III.

a) Distribution-level comparison with the survey reference: Table III places the simulated distributions next to the Carr-R respondent-sample reference under the estimand align-

ment of the reference protocol. The order-to-departure delay is measured identically on both sides—hours from the first delivered order, mandatory or voluntary, to the household’s first executed departure—and is pooled over the 114 departing order-recipient households across the three 24-household runs and the two 100-household runs of the scale demonstration below. The simulated CDF is directionally consistent but compressed: 0.570 of households depart immediately and 0.868 within thirty minutes, against 0.446 at both grid points in the survey ($n = 186$), and the survey’s long right tail (mean 7.1 h; 3.2% beyond 48 h) has no simulated counterpart. Two structural reasons bound the comparison: the 25-step scenario window caps observable delays at 12.5 h, and the half-hour decision cadence resolves each order at the next decision point, so sub-step hesitation and multi-day delays are outside the modeled behavior space. Survey channel prevalence is a multi-select estimand among warned respondents, so we report only like-for-like prevalence among order recipients rather than first-heard shares or distribution distances. All 197 simulated recipient households receive the official direct channel by cohort construction, and 0.939 also receive at least one interpersonal (community or intra-household) message, against survey interpersonal prevalence 0.372 (Wilson 95% CI [0.312, 0.435]); the survey splits official receipt across concrete channels (reverse-911 0.359, text 0.380), a taxonomy the scenario collapses into one direct channel. At the outcome level, the simulated individual departure share is 0.375 over all members and 0.458 within order-recipient households, well below the survey evacuation proportion 0.888 (Wilson 95% CI [0.849, 0.918], $n = 330$): unwarned simulated households rarely self-evacuate, and split parties leave members behind, whereas the survey estimand is an individual self-report and the non-evacuator reference group contains only 37 respondents. The study therefore supports Carr-informed process reasonableness rather than behavioral validation.

b) Scale demonstration: The same protocol is instantiated once more at 100 households—201 members, 153 of them decision-capable residents, receiving 86 official orders—for two seeds (101, 202) with the same decision model. Both runs complete all 25 steps successfully. Process behavior carries over qualitatively: complete safe household departure is 0.470 and 0.420 (against 0.514 ± 0.024 at 24 households), coordinated multi-adult departure 0.230 and 0.220, and split

TABLE III

DISTRIBUTION-LEVEL COMPARISON AGAINST THE CARR-R RESPONDENT-SAMPLE REFERENCE. DELAYS ARE HOURS FROM THE FIRST DELIVERED ORDER TO THE FIRST EXECUTED HOUSEHOLD DEPARTURE (SIMULATED: 114 DEPARTING ORDER-RECIPIENT HOUSEHOLDS POOLED OVER FIVE RUNS; SURVEY: $n = 186$). CHANNEL PREVALENCE IS MULTI-SELECT AMONG ORDER RECIPIENTS (SIMULATED: 197 HOUSEHOLDS; SURVEY: $n = 234$); FIRST-HEARD AND DISTRIBUTION-DISTANCE COMPARISONS LACK A MATCHED SURVEY ESTIMAND AND ARE NOT COMPUTED.

Quantity (grid point)	Simulated	Survey reference
<i>Order-to-departure delay CDF</i>		
$F(0\text{ h})$	0.570	0.446
$F(0.5\text{ h})$	0.868	0.446
$F(1\text{ h})$	0.965	0.634
$F(2\text{ h})$	1.000	0.656
$F(5\text{ h})$	1.000	0.774
$F(12\text{ h})$	1.000	0.892
Median / mean (h)	0.0/0.28	1.0/7.11
<i>Channel prevalence among order recipients</i>		
Official direct	1.000 [†]	—
Any interpersonal	0.939	0.372
(survey: reverse-911 / text)	—	0.359/0.380
<i>Individual departure share</i>		
All members	0.375	0.888
Order-recipient households	0.458	—

[†]By cohort construction; the survey splits official receipt across concrete channels not represented in the scenario.

departures remain the minority pattern. The message funnel scales with the population (2,240 and 2,119 messages sent) while preserving every life-cycle stage, and departures concentrate before the step-9 primary-route closure, with the 11 and 17 later parties rerouting to the alternate route. Arbitration remains contract-clean at scale: no executed party violates a care, vehicle, capacity, or caregiver contract in either run, and the 5 and 6 rejected departure requests (all for missing traveler intent) are recorded as rejections rather than executed violations. Within this demonstrated 100-household envelope we make no larger-size stability or throughput claim.

C. Which Mechanisms Enable Adaptation After Route Disruption?

This study supplies the central mechanism evidence. The controlled-scenario protocol instantiates the controlled-mechanism adapter with 8 households—each containing exactly two decision-capable adults, at least one dependent, and exactly one vehicle—over 10 steps with a controlled primary-route closure. Five conditions (the full configuration and four single-module removals; memory is the bounded warning-retention operationalization of Section IV-A1) crossed with 12 seeds give 60 planned cells; all 60 completed. The decision model is DeepSeek-V4-Flash at high thinking effort, and no call fails, falls back, or is served from cache. Opportunity is universal for the gating checks: every cell records all 8 households with a warning-exposed coordinator and all 16 decision-capable members holding a pre-closure primary-route plan, so no manipulation check is credit-blind.

Mechanism function versus downstream necessity. Table IV separates the two questions the ablation is designed to distinguish. All four modules pass their interface-level manipulation

TABLE IV

ABLATION RESULTS OVER 12 SEED PAIRS, SEPARATING MECHANISM FUNCTION (MANIPULATION CHECKS) FROM DOWNSTREAM NECESSITY (CONSEQUENCES OF REMOVAL). $\bar{\Delta}$ IS THE MEAN PAIRED FULL-MINUS-REMOVAL DIFFERENCE; THE SEEDS COLUMN COUNTS PAIRS WHOSE DIFFERENCE LIES IN THE PREDECLARED DIRECTION. A NEGATIVE $\bar{\Delta}$ FOR CLOSED-ROUTE REJECTIONS MEANS THE FULL CONDITION IS REJECTED LESS OFTEN. THE MEMORY CHECK SATURATES AT ITS STRUCTURAL CEILING WITH ZERO VARIANCE; IT VERIFIES THAT THE SWITCH CONTROLS THE INTENDED SIGNAL RATHER THAN ESTIMATING A GRADED BEHAVIORAL EFFECT.

Module	Quantity	$\bar{\Delta}$	Seeds
<i>Manipulation checks (interface function)</i>			
Memory	Warning retention	1.000	12/12
Feedback	Alternate-route update	0.964	12/12
Planning	Post-closure replan	0.906	12/12
Interaction	Feasible commitment	0.938	12/12
<i>Downstream consequences (necessity)</i>			
Feedback	Closed-route rej./hh	−6.542	12/12
Interaction	Coord. departure	0.812	12/12
Memory	Safe hh. departure	−0.104	2/12
Planning	Safe hh. departure	0.010	5/12

checks, each in the predeclared direction for 12/12 seed pairs (Fig. 4(a)): memory warning-retention saturates at its structural ceiling, and the feedback, planning, and interaction checks estimate genuine behavioral rates of 0.964, 0.906, and 0.938. Only two of the four modules, however, carry a determined downstream effect (Fig. 4(b,c)). Removing feedback raises closed-route rejections by 6.542 per household (12/12 seed pairs): the feedback switch jointly removes route-state refresh and bounded execution-failure feedback (Section IV-A1), so the private route view stays at its pre-closure state and residents repeatedly submit intentions the world must reject. Removing interaction lowers the coordinated-departure rate by 0.812 (12/12), and the condition levels give this contrast its precise meaning (Table V): half of the decision-capable residents still evacuate and dependent safety remains 1.000, while execution rejections rise to 3.833 per household. The interaction mechanism therefore governs the transition from individual intentions to household-level coordinated action, rather than evacuation capability itself. In contrast, the memory and planning contrasts for complete safe household departure are not determined: −0.104 with 2/12 seed pairs in the predicted direction and 25% exactly zero, and 0.010 with 5/12 and 17% exactly zero. The ablation therefore separates mechanism function from downstream necessity: all four mechanisms are causally active at their own interfaces, but in an 8-household, 10-step scenario with a single route closure, only feedback and interaction produce a measurable downstream signature. An undetermined downstream contrast is not evidence that a module is useless; it bounds what this scenario can attribute to it.

Condition levels. Table V reports descriptive condition levels (means over the 12 cells per condition) as context for the paired contrasts; they are not themselves predeclared contrasts. Three features stand out. First, removing feedback does not merely add rejections: complete safe household departure collapses from 0.812 to 0.021 and dependent safety from 1.000 to 0.055, because households depart on the closed primary

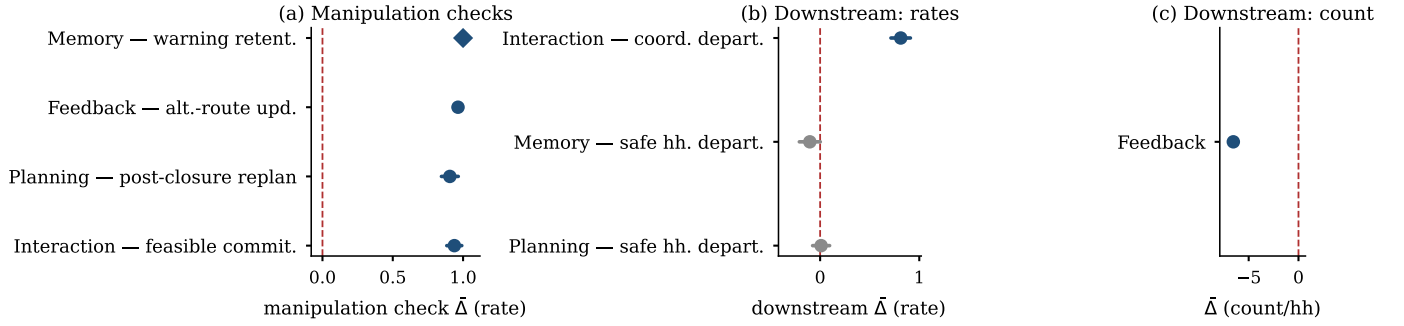


Fig. 4. Paired full-minus-removal contrasts in the ablation study over 12 seed pairs (points: mean paired difference $\bar{\Delta}$; bars: 95% paired- t intervals). (a) Manipulation checks; the memory check saturates at 1.000 with zero variance (diamond). (b) Downstream rate contrasts; gray marks contrasts crossing zero. (c) The downstream count contrast (closed-route rejections per household) on its own scale.

TABLE V

DESCRIPTIVE CONDITION LEVELS IN THE ABLATION STUDY (MEAN $\pm$ SD OVER THE 12 CELLS OF EACH CONDITION). SAFE HH. DEP. IS THE COMPLETE-HOUSEHOLD SAFE-DEPARTURE RATE, WHICH COINCIDES CELL BY CELL WITH THE COORDINATED-DEPARTURE RATE IN THIS SCENARIO; RESIDENT EVAC. IS THE DECISION-RESIDENT EVACUATION RATE; REJ./HH DENOTES REJECTIONS PER HOUSEHOLD. LEVELS ARE DESCRIPTIVE CONTEXT FOR THE PAIRED CONTRASTS OF TABLE IV, NOT PREDECLARED CONTRASTS THEMSELVES.

Condition	Safe hh. dep.	Resident evac.	Dependent safety	Closed-route rej./hh	Execution rej./hh
Full	0.812 $\pm$ 0.155	0.906 $\pm$ 0.078	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.198 $\pm$ 0.435
w/o memory	0.917 $\pm$ 0.123	0.958 $\pm$ 0.062	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.073 $\pm$ 0.241
w/o feedback	0.021 $\pm$ 0.072	0.042 $\pm$ 0.072	0.055 $\pm$ 0.075	6.542 $\pm$ 0.671	9.948 $\pm$ 0.880
w/o planning	0.802 $\pm$ 0.135	0.901 $\pm$ 0.068	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.396 $\pm$ 0.455
w/o interaction	0.000 $\pm$ 0.000	0.500 $\pm$ 0.000	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	3.833 $\pm$ 0.507
Rule reference [†]	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000

[†]Deterministic PADM-style reference policy on the full condition (same 12 seeds, zero LLM calls): a non-generative policy-class reference arm, not an ablation condition and not mechanism evidence.

route, accumulate execution rejections (9.948 per household), and never record an alternate-route update. Second, removing interaction eliminates coordinated multi-adult departures entirely (0.000 versus 0.812) while exactly half of the decision-capable residents still evacuate (0.500 with zero variance): the failure is one of coordination form, not of warning receipt or route knowledge, both of which remain near ceiling. Third, complete-household safe departure and coordinated departure coincide cell by cell in this scenario, so the two rates measure one process here even though the platform tracks them as distinct quantities. Memory and planning removal leave levels close to full (0.917 and 0.802 versus 0.812), consistent with their crossing-zero paired contrasts.

Rule-based reference arm. To answer what the same scenario produces without the generative decision model, a deterministic PADM-style reference policy—warning receipt triggers proposal exchange, a protective-action commitment, and feedback-driven rerouting to the open alternate route—was run through the identical kernel, message, and arbitration pipeline on the full condition for the same 12 seeds, with zero LLM calls and zero cost. The reference arm reaches the scenario ceiling on every seed (Table V), absorbing one transient route-capacity rejection per household before retrying on a later step. Two readings follow. First, the scenario is feasible to ceiling, so the generative full-condition shortfall (0.812 versus 1.000) and the removal collapses above reflect decision-process degradation rather than scenario hardness. Second, the ceiling is reached by a policy with exact structured state access and no language-understanding burden; the arm

bounds what a hand-coded policy can achieve and is a policy-class reference, not mechanism evidence.

D. Do the Mechanism Effects Replicate Across Models and Prompts?

This study tests whether the two determined ablation effects are artifacts of a single decision model. Seven models spanning different families and sizes (Table VI) each run three conditions—full, full-minus-feedback, and full-minus-interaction—over the same controlled scenario and prompt on seeds 2107, 2209, 2303, and 2411. Every model completes its 12 cells, giving 28 model-seed pairs pooled; the design is bounded at four seeds per model and is fully executed.

Table VI reports the replication in terms of condition levels rather than per-model effect intervals, because the question is whether each effect’s *direction* survives a change of decision model, not which model scores highest. The feedback effect replicates in direction for every model (Fig. 5(a)): under the full condition no model records any closed-route rejection, and removing feedback raises rejections per household in 4/4 seed pairs for all seven models (from 1.69 for GPT-5.4-mini up to 9.41 for Qwen3.6-27B). The interaction effect replicates in direction for six of seven (Fig. 5(b)), with removal driving coordinated multi-adult departure to zero in 4/4 seed pairs for six models. The exception is GPT-5.4-mini (1/4 seed pairs), which runs near the floor in the full condition (complete safe household departure 0.031, against 0.375–0.781 for the other six models), so little headroom remains for either removal effect to register a large drop; we read its interaction contrast

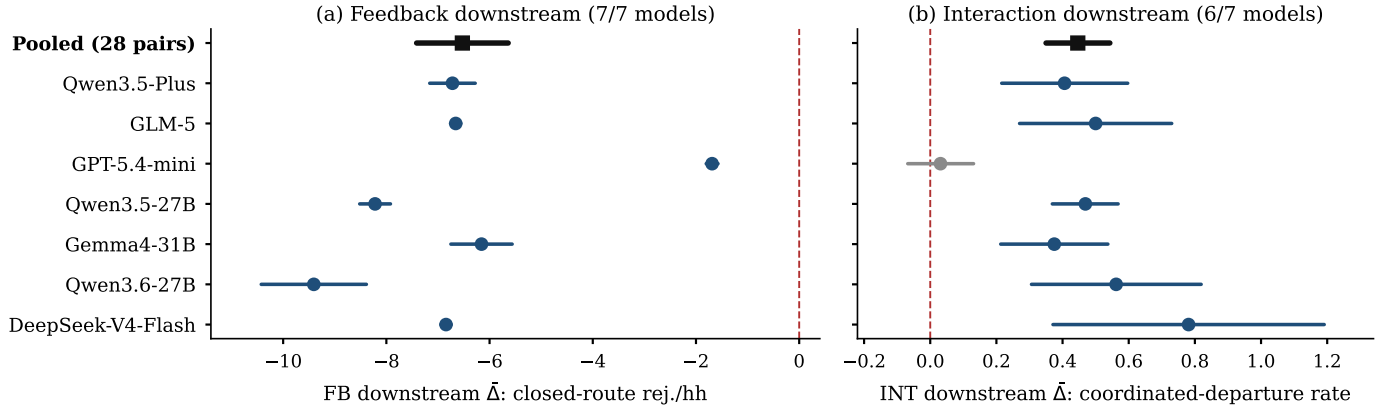


Fig. 5. Cross-model replication of the two determined ablation effects (per model: 4 seed pairs; pooled: 28 model–seed pairs; points and bars as in Fig. 4). (a) The feedback downstream effect replicates in direction for all seven models. (b) The interaction downstream effect replicates for six of seven; GPT-5.4-mini (gray) is the exception.

TABLE VI
CROSS-MODEL REPLICATION OF THE TWO DETERMINED MECHANISM EFFECTS OVER FOUR SEEDS PER MODEL. FB DENOTES CLOSED-ROUTE REJECTIONS PER HOUSEHOLD; INT DENOTES COORDINATED-DEPARTURE RATE. DIRECTION REPORTS THE SEED PAIRS (OF 4) IN THE PREDECLARED DIRECTION.

Decision model	Full safe hh. dep.	FB: closed rej./hh (full $\rightarrow$ -FB)	INT: coord. dep. (full $\rightarrow$ -INT)	Direction (FB · INT)
Qwen3.5-Plus	0.406	0.00 $\rightarrow$ 6.72	0.406 $\rightarrow$ 0.000	4/4 · 4/4
GLM-5	0.500	0.00 $\rightarrow$ 6.66	0.500 $\rightarrow$ 0.000	4/4 · 4/4
GPT-5.4-mini	0.031	0.00 $\rightarrow$ 1.69	0.031 $\rightarrow$ 0.000	4/4 · 1/4
Qwen3.5-27B	0.469	0.00 $\rightarrow$ 8.22	0.469 $\rightarrow$ 0.000	4/4 · 4/4
Gemma4-31B	0.375	0.00 $\rightarrow$ 6.16	0.375 $\rightarrow$ 0.000	4/4 · 4/4
Qwen3.6-27B	0.562	0.00 $\rightarrow$ 9.41	0.562 $\rightarrow$ 0.000	4/4 · 4/4
DeepSeek-V4-Flash	0.781	0.00 $\rightarrow$ 6.84	0.781 $\rightarrow$ 0.000	4/4 · 4/4

as a floor-limited estimand rather than as evidence against the direction. Full-condition process levels also vary widely (decision-resident evacuation 0.250–0.891), so the cross-model study supports the *signs*, not the magnitudes, of the two mechanism effects.

a) Prompt-wording sensitivity: A second robustness check varies prompt wording instead of the model. Two paraphrases of the controlled-scenario prompt—identical state JSON, schema, mechanism switches, and action set, with only the instruction wording changed—each run the same four seeds crossed with the same three conditions (24 cells in all) on DeepSeek-V4-Flash at temperature 0. Both paraphrases reproduce both determined effects on every seed pair: the feedback downstream contrast averages -7.000 and -7.094 closed-route rejections per household under the two variants, against -6.844 under the original wording (4/4 seeds in the predicted direction in all three wordings), and the interaction downstream contrast averages 0.938 and 0.875 coordinated multi-adult departure, against 0.781 (again 4/4 throughout). Manipulation checks pass in all 24 cells. This check bounds wording sensitivity within one model family; it is not evidence of prompt invariance across models or providers.

V. DISCUSSION

Mechanism function versus downstream necessity. The central scientific finding is the dissociation between the two: all four cognition and communication modules pass their

runner-aligned checks, yet only feedback and interaction carry determined downstream effects in the controlled scenario, and the condition levels show them playing different roles—feedback removal is the only condition that makes dependents unsafe (0.055 versus 1.000), because the household walks onto the closed route and cannot reroute, whereas interaction removal leaves every dependent safe but eliminates the coordinated form. Read at the domain level, the two determined channels are exactly the ones that cross a state-ownership boundary: feedback reconnects world-state changes and execution outcomes to private belief, and interaction converts private intentions into household coordination. The undetermined memory and planning contrasts admit several readings—bounded opportunity in a single-closure, 10-step scenario, a structural rather than graded memory check, or partial substitution that only a full factorial design could detect. The result suggests a broader evaluation lesson for other generative-agent disaster systems [12], [13], [14], [15]: a mechanism switch validated only at its own interface can overstate necessity if no downstream estimand is reported; each switch needs both a manipulation check and a downstream estimand with an explicit opportunity denominator. Throughout, the platform’s contracts held under full model-driven load, with zero arbitration violations in every reported run.

Model dependence. The directions of both determined effects replicate across seven models of different families

and sizes—feedback in all seven, interaction downstream in six—while magnitudes vary more than fivefold for feedback downstream, and full-condition process levels vary more still. Mechanism-level conclusions about signs are therefore not an artifact of the ablation’s decision model, but per-model quantitative claims remain model-dependent.

Boundaries and threats to validity. All results come from one Carr-informed controlled scenario archetype—one hazard trajectory, one controlled closure, 8 households in the ablation and cross-model studies, 24 in the process study, and 100 in the two-seed scale demonstration—and do not transfer to other hazards, scales, or network structures without new runs. Carr-R references are directional only: the estimands differ, the non-evacuator group contains 37 respondents, and both the delay CDF and the outcome level diverge from the survey (Table III), so neither is claimed to be behaviorally validated. The Carr-R test split has been opened for a feature–outcome diagnostic and no longer serves as an analyst-blind holdout. No blinded human or expert plausibility review has been completed, and scenario outputs remain computational artifacts rather than representations of real persons.

VI. CONCLUSION

We presented DisasterSociety, a generative agent-based simulation framework that separates private resident cognition, household-shared constraints and commitments, and authoritative world state, connecting them through an event-driven kernel in which the language model proposes but never mutates shared or physical state. In a Carr-informed controlled scenario, process studies at 24 and 100 households completed all runs with zero constraint violations; a paired twelve-seed ablation showed that memory, feedback, planning, and interaction each act at their own interfaces, while only feedback and interaction carry determined downstream effects; and a seven-model replication with prompt paraphrases reproduced both effect directions, with one model exception. Ongoing work extends cohort scale, hazard archetypes, and the cross-model seed count, and adds a prospective holdout and blinded expert judgment as independent evaluation channels.

REFERENCES

- [1] M. K. Lindell and R. W. Perry, “The protective action decision model: Theoretical modifications and additional evidence,” *Risk Analysis*, vol. 32, no. 4, pp. 616–632, 2012.
- [2] D. S. Mileti and P. W. O’Brien, “Warnings during disaster: Normalizing communicated risk,” *Social Problems*, vol. 39, no. 1, pp. 40–57, 1992.
- [3] S.-K. Huang, M. K. Lindell, C. S. Prater, H.-C. Wu, and L. K. Siebeneck, “Household evacuation decision making in response to hurricane ike,” *Natural Hazards Review*, vol. 13, no. 4, pp. 283–296, 2012.
- [4] E. Kuligowski, “Evacuation decision-making and behavior in wildfires: Past research, current challenges and a future research agenda,” *Fire Safety Journal*, vol. 120, p. 103129, 2021.
- [5] S. Grajdura, X. Qian, and D. Niemeier, “Awareness, departure, and preparation time in no-notice wildfire evacuations,” *Safety Science*, vol. 139, p. 105258, 2021.
- [6] S. Liu, P. M. Murray-Tuite, and L. Schweitzer, “Uniting multi-adult households during emergency evacuation planning,” *Disasters*, vol. 38, no. 3, pp. 587–609, 2014.
- [7] P. Maghelal, W. G. Peacock, and X. Li, “Evacuating together or separately: Factors influencing split evacuations prior to hurricane rita,” *Natural Hazards Review*, vol. 18, no. 2, p. 04016008, 2017.
- [8] J. S. Park, J. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, “Generative agents: Interactive simulacra of human behavior,” in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023, pp. 1–22.
- [9] J. Zhang, Y. Yan, J. Yan, Z. Zheng, J. Piao, D. Jin, and Y. Li, “A parallelized framework for simulating large-scale LLM agents with realistic environments and interactions,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*, 2025, pp. 1339–1349.
- [10] N. Bougie and N. Watanabe, “CitySim: Modeling urban behaviors and city dynamics with large-scale LLM-driven agent simulation,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 2025, pp. 215–229.
- [11] X. Zhou, Y. Sun, H. Yao, A. He, Y. Zhang, and W. Liu, “GASim: A graph-accelerated hybrid framework for social simulation,” in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2026, pp. 12 510–12 528.
- [12] R. Chen, C. Wang, Y. Sun, X. Zhao, and S. Xu, “From perceptions to decisions: Wildfire evacuation decision prediction with behavioral theory-informed LLMs,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 29 754–29 778.
- [13] P. Dang, J. Zhu, W. Li, Y. Xie, and H. Zhang, “Large-language-model-driven agents for fire evacuation simulation in a cellular automata environment,” *Safety Science*, vol. 191, p. 106935, 2025.
- [14] S. Yang, L. Ceferino, Y. Zhang, C. Gu, T. Guo, and G. Kondo, “When agents learn to think: Large language model-enhanced agent-based modeling for crowd evacuation in disaster scenarios,” *Reliability Engineering & System Safety*, vol. 269, p. 112056, 2026.
- [15] R. Sultimov, M. Mozikov, D. Abramov, M. Kovalchuk, M. Malykh, I. Makarov, A. Osipov, A. Volkov, and Y. Maximov, “RESPOND: Realistic environment simulation of population and natural disasters with LLM-driven agents,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 48, pp. 41 694–41 696, 2026, demonstration Track.
- [16] L. Palen and A. L. Hughes, “Crisis informatics: Human-centered research on tech & crises,” *Annual Review of Information Science and Technology*, vol. 52, no. 1, pp. 163–185, 2018.
- [17] P. Murray-Tuite and B. Wolshon, “Evacuation transportation modeling: An overview of research, development, and practice,” *Transportation Research Part C: Emerging Technologies*, vol. 27, pp. 25–45, 2013.
- [18] R. J. Beckman, K. A. Baggerly, and M. D. McKay, “Creating synthetic baseline populations,” *Transportation Research Part A: Policy and Practice*, vol. 30, no. 6, pp. 415–429, 1996.
- [19] X. Ye, K. C. Konduri, R. M. Pendyala, B. Sana, and P. Waddell, “A methodology to match distributions of both household and person attributes in the generation of synthetic populations,” in *TRB 88th Annual Meeting Compendium of Papers DVD*. Transportation Research Board, 2009, pp. 1–24, paper 09-2096.
- [20] M. Nagarajan, D. Shaw, and P. Albores, “Disseminating a warning message to evacuate: A simulation study of the behaviour of neighbours,” *European Journal of Operational Research*, vol. 220, no. 3, pp. 810–819, 2012.
- [21] R. G. Sargent, “Verification and validation of simulation models,” *Journal of Simulation*, vol. 7, no. 1, pp. 12–24, 2013.
- [22] F. Klügl, “A validation methodology for agent-based simulations,” in *Proceedings of the 2008 ACM Symposium on Applied Computing*, ser. SAC ’08. Association for Computing Machinery, 2008, pp. 39–43.
- [23] V. Grimm, E. Revilla, U. Berger, F. Jeltsch, W. M. Mooij, S. F. Railsback, H.-H. Thulke, J. Weiner, T. Wiegand, and D. L. DeAngelis, “Pattern-oriented modeling of agent-based complex systems: Lessons from ecology,” *Science*, vol. 310, no. 5750, pp. 987–991, 2005.
- [24] V. Grimm, S. F. Railsback, C. E. Vincenot, U. Berger, C. Gallagher, D. L. DeAngelis, B. Edmonds, J. Ge, J. Giske, J. Groeneveld, A. S. Johnston, A. Milles, J. Nabe-Nielsen, J. G. Polhill, V. Radchuk, M.-S. Rohwäder, R. A. Stillman, J. C. Thiele, and D. Ayllón, “The ODD protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism,” *Journal of Artificial Societies and Social Simulation*, vol. 23, no. 2, p. 7, 2020.
- [25] H. Zhao, Z. Liang, T. Ma, X. Shi, M. Kapadia, T. Thrash, C. Hoelscher, J. Jia, B. Liu, and J. Cao, “Adversarial reinforcement learning for enhanced decision-making of evacuation guidance robots in intelligent fire scenarios,” *IEEE Transactions on Computational Social Systems*, vol. 12, no. 5, pp. 2030–2046, 2025.
- [26] C. A. Bail, “Can generative ai improve social science?” *Proceedings of the National Academy of Sciences*, vol. 121, no. 21, p. e2314021121, 2024.

- [27] A. N. N. Rende, T. Yilmaz, and O. Ulusoy, “Crowd: A social network simulation framework,” *IEEE Transactions on Computational Social Systems*, vol. 12, no. 6, pp. 4369–4383, 2025.
- [28] X. Guo, K. Huang, J. Liu, W. Fan, N. Vélez, Q. Wu, H. Wang, T. L. Griffiths, and M. Wang, “Embodied LLM agents learn to cooperate in organized teams,” *IEEE Transactions on Computational Social Systems*, vol. 13, no. 2, pp. 2514–2530, 2026.
- [29] M. G. Mendoza, L. Waldburger, J. Lee, and S. Sastry, “Hierarchical generative agents for simulating sequential human behavior,” 2026, preprint.